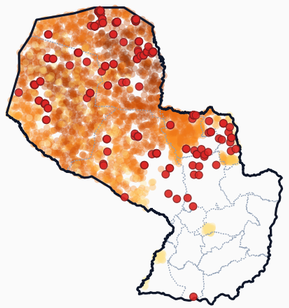
INCENDIOS_PY · SOVEREIGN GEOAI SYSTEM

Decoupled Biophysical Ignition Hazard & Tactical Wildfire Propagation Framework

24-HOUR FORECAST · 1-km² RESOLUTION · SUB-SECOND INFERENCE

PHASE 1 Multisource Ingestion

Paraguay (1 km² Grid)
VIIRS Fire Hotspots



NASA FIRMS (VIIRS 375m)

Suomi-NPP, NOAA-20, NOAA-21
Bow-Tie filtered (scan ≤ 1.5 km)

ERA5-Land Reanalysis

0.25° grid (640 national nodes)
VPD, Wind, CDD, FFMC, ISI

MapBiomas Collection 4

30m aggregated to 1 km²
5 dominant fuel/forest classes

STAGE 1: KINEMATIC BIO-PHYSICAL GATING

Condition: VPD ≥ 0.8 kPa AND Rainfall 48h < 10 mm AND Fuel = 1

- Automatically excludes water, wet savannas & saturated air cells.
- Discards 92% of non-fire national territory with 0% false dismissal.
- Reduces computational space from 1.46B to 4.4M candidate cell-days.

Physical Drivers

Thermal Foci (t-1)

PHASE 2 Decoupled Dual-Layer GeoAI Engine

MODE 1: PURE BIOPHYSICAL IGNITION HAZARD

Features: Weather + Fuel Structure + Ecotone Edge Contrast

- STRICT CAUSAL ISOLATION: No fire history lags allowed (t-1 = 0).
- Drivers: VPD max, Thermal amplitude, Wind speed, FFMC, Ecotone.
- Calibrated via Monotonic Bounded Platt Scaling (w in [0.1, 10.0]).

Output: P(Ignition) --> Pre-suppression & Forest Patrol

MODE 2: TACTICAL FIRE SPREAD & CONTAGION

Features: Physical Drivers + Spatial Proximity Kernel (t-1)

- Causal spatial memory: dist_foco_t1_km, focos_25km_t1, frp_25km_t1.
- Models convective advection & active flame front advancement.
- Evaluated with 20-km Dead-Zone spatial buffer to prevent leakage.

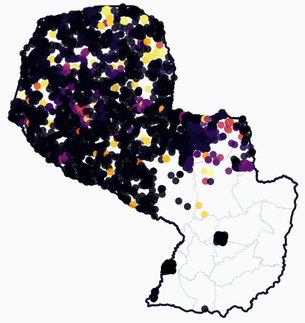
Output: P(Propagation) --> Tactical Firefighting Deployment

P(Ignition) Forecast

P(Spread) Validation

PHASE 3 Sovereign Alert & Ben

Operational Alert Map
(1 km² Resolution)



OPERATIONAL METRICS

• Inference Latency:	< 0.75 s
• Sovereign Grid:	400,006 cells
• Tactical Precision:	97.71 %
• Chaco Spread ROC:	0.941
• Oriental Ignition ROC:	0.771

International Benchmarking on Blind Out-of-Time Test Set

